Intuitive Terrain Difficulty Scaling Plan

This plan redesigns the procedural generation math for all 11 RS terrains so that the `difficulty` parameter (0.0 to 1.0) maps cleanly and intuitively to percentages of the robot's physical capabilities.

> [!TIP] **Go2 Reference Scale:**
>
> - `body_length` ≈ 0.75m | `standing_height` ≈ 0.30m | `limb_reach` ≈ 0.43m | `foot_length` ≈ 0.04m

Redesigned Concepts (Based on your feedback!)

RS5: Stairs (Testing Staircase Cadence/Endurance)

You made a great point. RS10 (High Step) already tests pure vertical climbing height. RS5 (Stairs) should test the robot's ability to coordinate a continuous, rhythmic climb over a long flight of stairs!

- **Scales with Difficulty:**
  - `step_count = 3 + round(10 * difficulty)` *(Flight gets longer as difficulty increases!)*
  - **Easy (0.25):** 6 steps
  - **Medium (0.50):** 8 steps
  - **Hard (0.75):** 11 steps
- **Fixed:**
  - `step_rise_m = standing_height * 0.4` *(Fixed at 40% standing height, ~12cm for Go2, which is a perfect standard stair)*
  - `step_tread_m = max(foot_length * 2.5, body_length * 0.35)` *(Fixed at ~26cm)*

RS6: Repeated Waves (Testing Spatial Frequency)

Instead of scaling the amplitude (height) of the waves, we can scale how tightly packed the waves are! We can change the math so difficulty adds more waves into the same fixed length.

- **Scales with Difficulty:**
  - `wave_count = 1 + round(4 * difficulty)` *(Number of full sine waves in the track)*
  - **Easy (0.25):** 2 full waves (Wide and gentle)
  - **Medium (0.50):** 3 full waves (Tighter)
  - **Hard (0.75):** 4 full waves (Very tight and aggressive undulating)
- **Fixed:**
  - `length_m = body_length * 5.0` *(Fixed at ~3.75m)*
  - `amplitude_m = standing_height * 0.3` *(Fixed at ~9cm high waves)*

- `segments = 30` *(Fixed high enough to keep them smooth)*

RS7: Curved Hill (Testing Jagged Polygon Transitions)

For a single hill, instead of just making it taller, we can change the `segments` variable to make the hill "blockier" and more jagged as difficulty increases!

- **Scales with Difficulty:**
  - `segments = max(4, 24 - round(20 * difficulty))` *(Fewer segments = sharper, blockier angles!)*
  - **Easy (0.25):** 19 segments (A very smooth, perfectly curved hill)
  - **Medium (0.50):** 14 segments (A slightly chunky hill)
  - **Hard (0.75):** 9 segments (A very jagged, polygonal hill that tests sharp foot contacts)
- **Fixed:**
  - `length_m = body_length * 5.0` *(Fixed at ~3.75m)*
  - `amplitude_m = standing_height * 0.7` *(Fixed at ~21cm tall)*

Unchanged Intuitive Plans (From Before)

RS1 & RS2: Slippery & Rough Flat Ground

- **Scales:** `roughness_amplitude_m = foot_length * difficulty` (Easy = 1cm, Med = 2cm, Hard = 3cm bumps).
- **Fixed:** `length_m = body_length * 5.0`

RS3 & RS4: Slopes (Smooth & Rough)

- **Scales:** `slope_rad = 45° * difficulty` (Easy = 11°, Med = 22.5°, Hard = 34°). RS4 Roughness scales up to 100% foot length.
- **Fixed:** `run_m = body_length * 5.0`

RS8: Small Repeated Gaps

- **Scales:** `gap_m = foot_length * (2.0 * difficulty)` (Easy = 2cm, Med = 4cm, Hard = 6cm gaps).
- **Fixed:** `gap_count` = 5 gaps, `platform_m` = ~22cm.

RS9: Large Gap

- **Scales:** `gap_m = body_length * difficulty` (Easy = 19cm, Med = 37.5cm, Hard = 56cm leap).
- **Fixed:** `takeoff_m = body_length`

RS10: High Step

- **Scales:** `rise_m = limb_reach * difficulty` (Easy = 11cm, Med = 21.5cm, Hard = 32cm step up).

- **Fixed:** `platform_m = body_length * 2.0`

RS11: Marsh Approximation (Mud)

- **Scales:** Mud firmness drops by 50% and stickiness (damping) increases by 100% as difficulty scales to Max.
- **Fixed:** `length_m = body_length * 5.0`

> [!IMPORTANT] How do these new concepts for Stairs (scaling number of stairs), Waves (scaling frequency), and Hills (scaling jaggedness) sound to you? This creates a completely unique physical test for every single terrain type!